Software Project Management Project

Semester Project, SS 2026

Group 1

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iGEO Internet of Moving Things Planning Report

Authors

Hieu Nguyen

Jingru Wang

Prasad Dinesha Shyamala

Jiahao Yi

Teng Sun

Ziyu Guo

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1: Project Management Foundations, Stakeholder Analysis, and Decision Analysis

Project Definition

TOMI Automobiles plans to extend the GEO car line with iGEO, a decoupled Internet of Moving Things (IoMT) platform. The project introduces dedicated IoMT hardware, a dedicated in-car display, a hosting application for software modules, and two demonstration modules: Drive to Welcoming Home (WH) and Smart Commercials (SC). The system is deliberately separated from core driving and safety-critical vehicle functions.

The planning objective of this report is to turn the project description into a manageable project plan that covers scope, stakeholders, supplier selection, work breakdown, effort estimation, schedule dependencies, and human resource planning.

Project Scope

- Design a modular IoMT platform for the GEO product line without influencing driving-critical behavior.
- Select suitable iGEO IoMT hardware and display components and justify the choices.
- Develop or outsource the hosting application and its GUI for module deployment, activation, deactivation, and configuration.
- Deliver two exemplary modules: WH for smart-home arrival automation and SC for location-based commercial notifications.
- Integrate iGEO Cloud, smart-home devices, garage devices, SMS notification, and retail cloud services through controlled interfaces.

Business Goals

- Strengthen TOMI's innovation image in the emerging Internet of Moving Things market.
- Increase the attractiveness of the GEO line for young, connected, and tech-oriented customers.
- Differentiate TOMI from slower-moving automotive competitors through software-enabled customer experience.
- Explore future monetization potential through contextual offers while avoiding intrusive user experience.
- Maintain customer trust by treating safety, security, and privacy as visible product qualities.

Project Goals

- Complete the iGEO project within the one-year window from 2023-06-01 to 2024-05-31.
- Keep total implementation cost below EUR 1,000,000.
- Keep all IoMT software and hardware decoupled from GEO driving-control systems.
- Provide a user-friendly display and GUI for module management and configuration.
- Demonstrate working WH and SC modules with security, privacy, integration, and usability validation.

Critical Success Factors

- Clear, validated requirements and strict scope control for the hosting platform and both modules.
- A supplier decision that balances safety, process maturity, IoT competence, cost, and collaboration.
- A modular architecture that supports future IoMT modules without redesigning the platform.
- Robust security/privacy design for location data, home access, garage access, and user profiles.
- Early stakeholder involvement, especially from Finance, IT, Architecture, Sales, and Product Line Management.

Assumptions and Constraints

Type	Item
Assumption	Users accept controlled use of location, profile, and smart-home data when convenience and privacy

	controls are clear.
Assumption	External IoT and retail services provide stable APIs or protocols for integration.
Assumption	The selected supplier can provide hardware/display support within the one-year schedule.
Constraint	Budget is capped at EUR 1,000,000.
Constraint	The project must finish within one year, from 2023-06-01 to 2024-05-31.
Constraint	IoMT features must not control or endanger driving-critical car functions.
Constraint	TOMI has limited internal capacity, so hardware/display and hosting application work are partly outsourced.

Stakeholders

Stakeholder	Role	Attitude	Reasoning
George Wein	Project Manager	Supportive	Initiated the idea and owns coordination after project approval.
Tom Wards	CTO	Supportive but cautious	Wants innovation but insists on security, safety, testing, and clear requirements.
Jim Becker	Executive Marketing	Supportive with price concern	Sees campaign potential but worries about affordability.
Julia Andersen	Finance and Controlling Director	Skeptical	Concerned about development cost, profit per unit, and financial risk.
Peter Scofield	Executive IT	Skeptical	Concerned about security, privacy, media risk, and IT capacity.
Anne Smith	Lead IT Architect	Hidden supportive	Believes separation architecture can work but avoids conflict with Peter.
Jane Wilson	Sales Manager	Interested	Needs customer segments and target regions for sales planning.
Jim Bucksley	Developer	Supportive	Likes the concept and sees lower risk because core car functions are untouched.
David Renegan	Developer	Cautiously supportive	Likes the idea but knows requirements will be difficult.
John Burton	GEO Product Line Manager	Supportive	Sees alignment with the GEO product strategy.
Jonathan Newman	TEO Product Line Manager	Hidden opposed	Privately believes the features distract from the driving experience and create requirement risk.

Socio-Diagram and Stakeholder Relations

The socio-diagram focuses on attitudes and informal influence. Green relations indicate support or constructive cooperation, yellow relations indicate caution or political pressure, and red/dashed relations indicate open or hidden resistance.

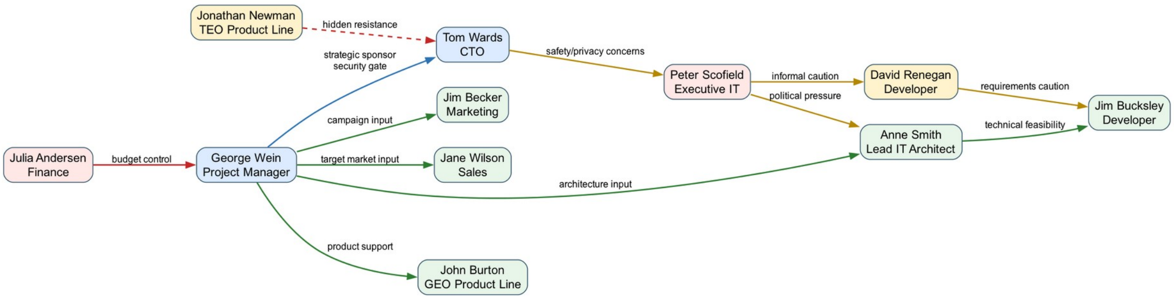

Figure 1: Socio-diagram of the iGEO stakeholder environment.

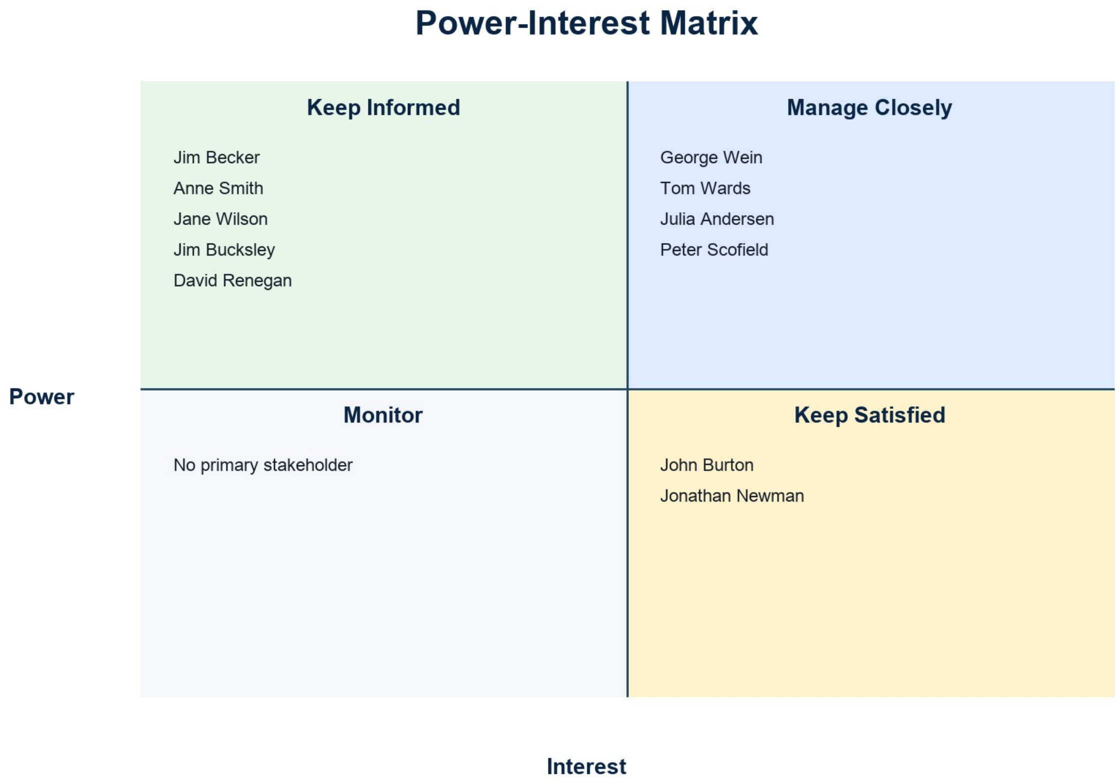

Figure 2: Power-interest matrix derived from stakeholder attitude and influence.

Stakeholder Involvement Counter Actions

Stakeholder Group	Risk	Counter Action
Finance and IT skeptics	Budget, security, privacy, and capacity concerns may slow approval.	Use monthly risk and cost reviews; show security architecture and cost forecast early.
Hidden opponents	Jonathan Newman may influence Tom informally against the project.	Address driving-experience concerns explicitly and frame iGEO as optional, decoupled, and non-safety-critical.
Technical contributors	Developers and architecture may underestimate requirement complexity.	Run structured requirement reviews and trace requirements to use cases and test cases.
Marketing and sales	Commercial value may remain vague.	Define target customer assumptions, price sensitivity, and user acceptance criteria before launch.
Supportive sponsors	Innovation enthusiasm may create scope creep.	Keep a controlled backlog and require change impact analysis for new modules or features.

Supplier Selection Criteria

The supplier criteria are limited to five decision criteria, as required by the task. They emphasize the concerns raised by TOMI stakeholders and the information available in the supplier profile.

Criterion	Rationale
Safety/security and reputation	The project handles location, home, garage, and personal profile data; reputation risk is repeatedly raised by CTO and IT.
Automotive/IoT technical capability	The supplier must support smart-car and IoT integration, including reliable hardware/display choices.
Process maturity and quality assurance	Testing, deployment, and change handling are key concerns for TOMI.
Cost and commercial fit	The project is capped at EUR 1,000,000 and affordability is a marketing concern.
Flexibility and collaboration	Requirements are likely to evolve, so supplier communication and change handling matter.

2: Work Breakdown Structure and Decision Analysis & Resolution

WBS First Version

The first WBS follows a broad functional decomposition. It is useful for orientation but still contains overlap and missing work that must be corrected.

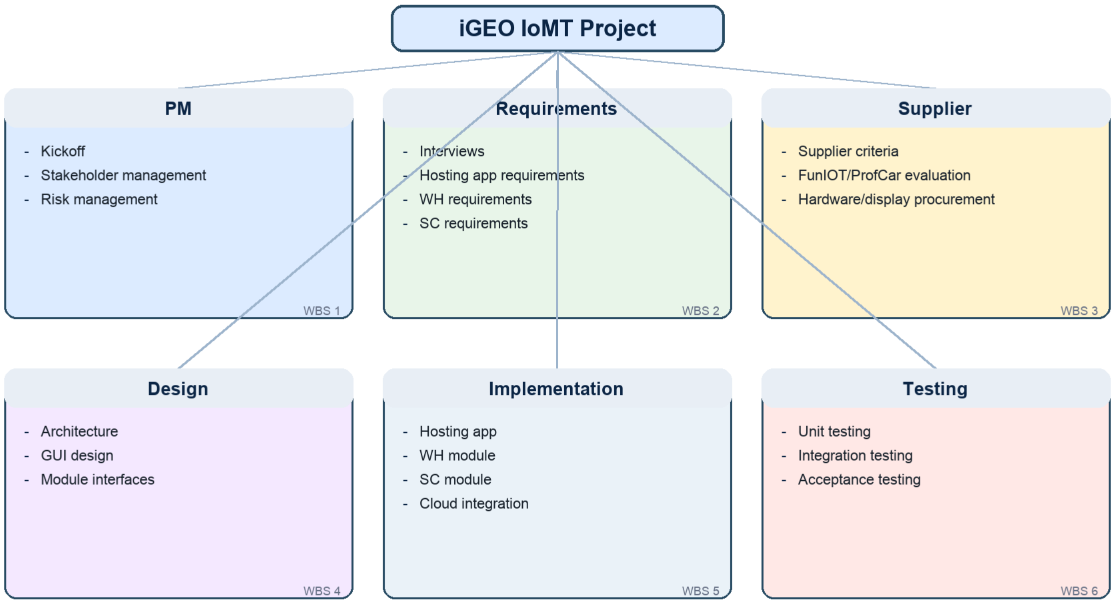

Figure 3: WBS version 1.

WBS Validation and Error Solving

Rule	Violation in Version 1	Resolution for Version 2
Disjunction Rule	Hardware/display procurement appeared under both supplier management and technical design.	Separate supplier decision/contracting from technical integration design.
Disjunction Rule	GUI design overlapped with WH and SC module work.	Move common GUI/UX responsibility to Hosting GUI and Module Configuration UI; keep WH/SC focused on module logic.
Completion Rule	Security/privacy validation was too implicit.	Add a dedicated verification work package for security, privacy, usability, and integration tests.
Completion Rule	Deployment and operational handover were missing.	Add a release, documentation, and handover work package.

WBS Final Version

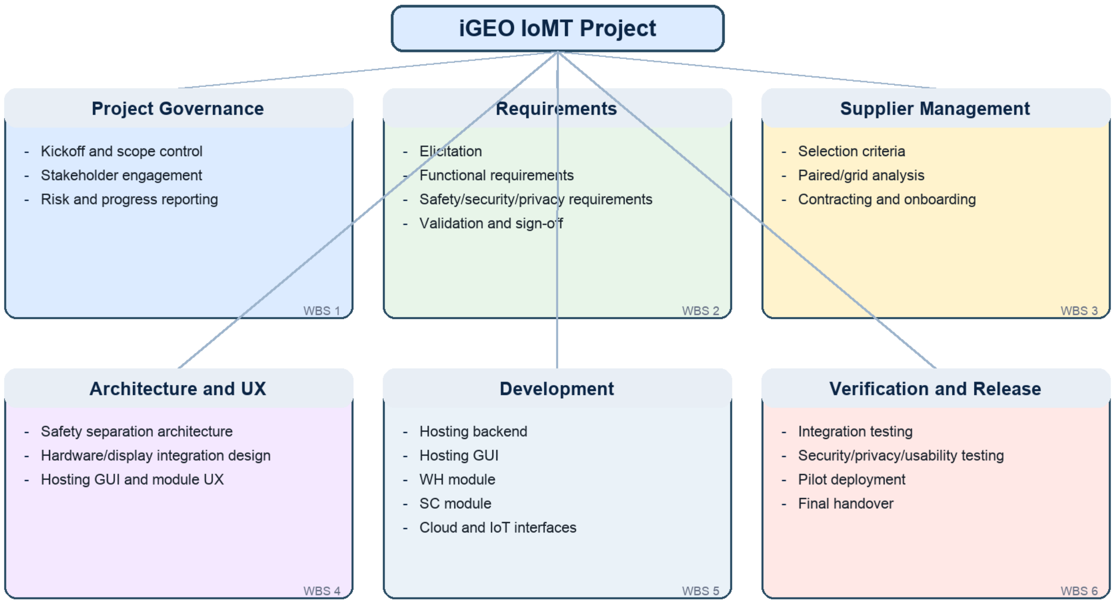

Figure 4: WBS version 2 after validation.

Final WBS Work Packages

ID	Work Package	Description
WP1	Project initiation and stakeholder management	Kick off the project, align key stakeholders, define governance and communication routines.
WP2	Requirements engineering	Elicit, document, prioritize, and validate functional and non-functional requirements.
WP3	Supplier evaluation and contracting	Define criteria, evaluate FunIOT/ProfCar, and onboard the chosen supplier.
WP4	Architecture and safety separation design	Design the decoupled IoMT architecture and integration boundaries.
WP5	Hardware/display selection and integration design	Select automotive-grade hardware/display options and define integration constraints.
WP6	Hosting application backend	Implement deployment, enable/disable, compatibility checks, and lifecycle control.
WP7	Hosting GUI and module configuration UI	Build user-facing GUI flows for module management and configuration.
WP8	WH module implementation	Implement smart-home, lighting, climatization, garage, and notification behavior.
WP9	SC module implementation	Implement commercial tag configuration, location-based offers, and notification display.
WP10	Cloud and third-party integration	Integrate iGEO cloud, smart home devices, garage, SMS, and retail cloud interfaces.
WP11	Verification, security, and usability testing	Run integration, acceptance, security/privacy, and display usability tests.
WP12	Deployment, documentation, and handover	Prepare pilot release, operational documentation, training, and final handover.

Supplier Decision

Paired Comparison Analysis

The paired comparison ranks the five criteria before the grid analysis. Safety/security and technical capability dominate because failure would threaten customer trust and the project's feasibility. Cost is important but cannot compensate for a risky automotive IoMT supplier.

Pair	Preferred Criterion	Reason
Safety/security vs Technical capability	Safety/security	Stakeholders repeatedly treat safety and privacy as approval gates.
Safety/security vs Process maturity	Safety/security	Process matters, but reputation damage from a breach would be more severe.
Safety/security vs Cost	Safety/security	A cheaper supplier is not acceptable if security risk increases.
Safety/security vs Flexibility	Safety/security	Flexibility cannot replace secure automotive-grade behavior.
Technical capability vs Process maturity	Technical capability	The system needs proven IoT/smart-car competence first.
Technical capability vs Cost	Technical capability	Integration failure would be more expensive than a higher day rate.
Technical capability vs Flexibility	Technical capability	Technical fit is more decisive than change style.
Process maturity vs Cost	Process maturity	CMMI/ITIL/ISO-like maturity supports predictable delivery and testing.
Process maturity vs Flexibility	Process maturity	Controlled deployment is more important than informal adaptability.
Cost vs Flexibility	Cost	Within a fixed budget, commercial fit receives the remaining priority.

Criterion	Rank Weight
Safety/security and reputation	5
Automotive/IoT technical capability	4
Process maturity and quality assurance	3
Cost and commercial fit	2
Flexibility and collaboration	1

Grid Analysis

Criterion	Weight	FunIOT Score	FunIOT Weighted	ProfCar Score	ProfCar Weighted
Safety/security and reputation	5	4	20	7	35
Automotive/IoT technical capability	4	6	24	9	36
Process maturity and quality assurance	3	4	12	9	27
Cost and commercial fit	2	9	18	4	8
Flexibility and collaboration	1	9	9	5	5
Total			83		111

Decision: ProfCar is selected with 111 weighted points compared with FunIOT's 83. FunIOT is attractive for cost and flexibility, but ProfCar better matches TOMI's security-sensitive automotive context due to stronger smart-car experience, larger engineering capacity, ISO certification, CMMI/ITIL experience, 5G capability, and longer warranty.

3: Planning Poker and Use Case Point Estimation Method

Planning Poker

The group used a Fibonacci-like scale of 1, 2, 3, 5, 8, 13, 20, and 40 points. The scale was chosen because the uncertainty between large work packages grows non-linearly and the gaps force discussion instead of false precision.

WP	Final Estimate	First-Round Range	Rationale
WP1	5	3-8	Mostly coordination work; uncertainty comes from stakeholder availability.
WP2	8	5-13	Requirements are described in the brief but stakeholder conflicts make validation harder.
WP3	8	5-13	Supplier information is available, but criteria weighting and contract risk require discussion.
WP4	13	8-20	Safety separation and modularity are central architectural concerns.
WP5	13	8-20	Hardware/display choice depends on supplier data and user-facing usability constraints.
WP6	20	13-40	The hosting app controls module lifecycle, compatibility checks, and deployment behavior.
WP7	13	8-20	GUI complexity is moderate but must remain safe and simple while driving.
WP8	20	8-40	WH touches smart home, garage, passenger identification, cancellation, and privacy flows.
WP9	13	8-20	SC is smaller than WH but still needs location, preference, and retail integration.
WP10	20	13-40	External APIs and device protocols increase uncertainty.
WP11	20	13-40	Testing is broad because security, privacy, usability, and integration all matter.
WP12	8	5-13	Release and handover are well understood but depend on prior acceptance results.

The hardest work package to estimate was WP8, the WH module. The first round ranged from 8 to 40 points because some members viewed it as ordinary feature coding, while others included passenger identification, smart-home device reliability, garage access, privacy, cancellation behavior, and error handling. The group selected 20 points after agreeing to treat WH as the most integration-heavy module.

The largest gap was therefore 32 points on WP8. WP10 and WP11 also produced large gaps because third-party integration and security/usability validation depend on supplier decisions and external device behavior.

Use Case Point Estimation

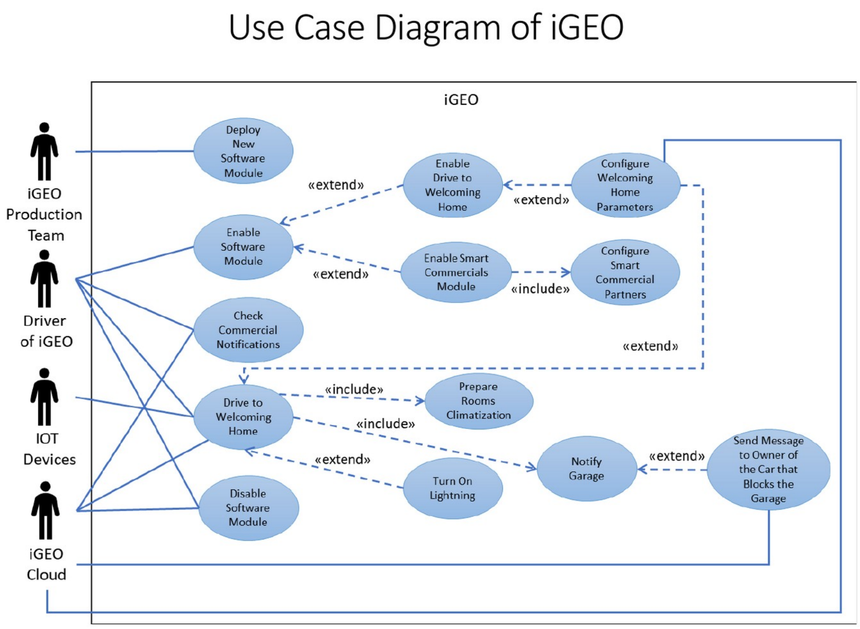

Figure 5: Provided use case diagram of iGEO.

Unadjusted Actor Weight (UAW)

Actor Type	Actors	Classification Rationale	Quantity	Weight	Subtotal
Simple	iGEO Cloud	Defined API used for profiles, configuration storage, and SMS forwarding.	1	1	1
Average	IoT Devices at Smart Home; IoT Garage Device	External systems communicate through network/device protocols.	2	2	4
Complex	Driver; iGEO Production Team	Human actors interact through graphical user interfaces.	2	3	6

Total UAW = 11. Supermarket and retail cloud services are mentioned as preconditions but are not counted separately because they are not primary actors in the provided use cases.

Unadjusted Use Case Weight (UUCW)

Use Case	Transactions	Type	Weight
Deploy New Software Module	4	Average	10
Enable Software Module	3	Simple	5
Enable Drive to Welcoming Home Module	3	Simple	5
Configure Welcoming Home Parameters	4	Average	10
Drive to Welcoming Home	5	Average	10
Prepare Rooms Climatization	5	Average	10
Turn On Lightning	5	Average	10
Notify Garage	5	Average	10
Send Message to Owner of Blocking Car	2	Simple	5
Enable Smart Commercial Module	3	Simple	5
Configure Smart Commercial Parameters	4	Average	10
Check Commercial Notification	3	Simple	5
Disable Software Module	6	Average	10

Use Case Type	Rule	Quantity	Weight	Subtotal
Simple	Up to 3 transactions	5	5	25
Average	4 to 7 transactions	8	10	80
Complex	More than 7 transactions	0	15	0

UUCW = 105. Therefore, UUCP = UAW + UUCW = 11 + 105 = 116.

Technical Complexity Factor

Factor	Description	Project Rationale	Weight	Score	Subtotal
T1	Distributed solution	Vehicle hardware, cloud services, IoT home devices, garage devices, and retailer services cooperate.	2	5	10
T2	Performance objectives	Garage and in-car notifications should react quickly and predictably.	1	4	4
T3	End-user efficiency	Drivers need short, clear interaction flows while using the car.	1	4	4
T4	Complex internal processing	The WH module combines passenger identification, location, climatization, lighting, and garage logic.	1	4	4
T5	Reusable code/design	The hosting app must support future modules beyond WH and SC.	1	4	4
T6	Easy to install	Supplier and workshop installation should be repeatable across GEO vehicles.	0.5	3	1.5
T7	Easy to use	Usability is critical because the dedicated display is used in a driving context.	0.5	5	2.5
T8	Portable	A reusable IoMT platform may later be transferred to other TOMI product lines.	2	3	6
T9	Easy to change	Module configuration, supplier components, and cloud APIs may change after release.	1	4	4
T10	Concurrent users	Multiple drivers and profiles may use the same vehicle or cloud account over time.	1	3	3
T11	Special security features	Location, garage access, home devices, and personal profiles require strong protection.	1	5	5
T12	Third-party access	Smart home, garage, SMS, and retail services require third-party interfaces.	1	4	4
T13	Special training facilities	The system should be intuitive; only minimal handover guidance is expected.	1	1	1

Technical Factor Total = 53.0. TCF = 0.6 + (0.01 x 53.0) = 1.13.

Environmental Complexity Factor

Factor	Description	Project Rationale	Weight	Impact	Subtotal
E1	Familiar with software development process	The team is familiar with structured SPM methods but not with every automotive process.	1.5	3	4.5
E2	Application experience	The team has limited direct automotive IoMT domain experience.	0.5	2	1.0
E3	Object-oriented paradigm experience	The implementation assumptions use familiar object-oriented design concepts.	1	4	4
E4	Lead analyst capability	The analyst role is considered capable for an academic planning project.	0.5	3	1.5
E5	Motivation	The project is visible and relevant for exam admission, so motivation is high.	1	5	5
E6	Stable requirements	The brief is clear, but stakeholder concerns and supplier choices create some volatility.	2	3	6
E7	Part-time workers	The actual team members are students and can contribute only part-time.	-1	4	-4
E8	Difficulty of programming language	Typical implementation languages and APIs are familiar enough for the assumed team.	-1	2	-2

Environmental Factor Total = 16.0. ECF = 1.4 + (-0.03 x 16.0) = 0.92.

Total Use Case Points: UCP = UUCP x TCF x ECF = 116 x 1.13 x 0.92 = 120.59.

Productivity factor: E1-E6 below 3 plus E7-E8 above 3 gives 2 risk factors, so PF = 20 person-hours/UCP. Estimated development effort = 120.59 x 20 = 2412 person-hours.

4: Project Scheduling

Activity Network Diagram and Critical Path

The activity network uses weeks as the time unit and includes all four precedence relationships: Finish-to-Start (FS), Start-to-Start (SS), Finish-to-Finish (FF), and Start-to-Finish (SF). The SF relationship is used for operational handover: support readiness cannot finish before pilot deployment preparation has started.

ID	Activity	Dur.	Predecessors	ES	EF	LS	LF	Float	Start	Finish
A1	Project initiation and stakeholder alignment	2	-	0	2	0	2	0	2023-06-01	2023-06-14
A2	Requirements engineering	6	A1 FS	2	8	2	8	0	2023-06-15	2023-07-26
A3	Supplier criteria, evaluation, and contracting	5	A2 SS	2	7	6	11	4	2023-06-15	2023-07-19
A4	Architecture and safety separation concept	5	A2 FS	8	13	8	13	0	2023-07-27	2023-08-30
A5	Hardware/display selection and integration planning	8	A3 FS	7	15	11	19	4	2023-07-20	2023-09-13
A6	Hosting application implementation	12	A4 FS	13	25	13	25	0	2023-08-31	2023-11-22
A7	WH module implementation	10	A6 SS	13	23	15	25	2	2023-08-31	2023-11-08
A8	SC module implementation	8	A6 SS	13	21	17	25	4	2023-08-31	2023-10-25
A9	Cloud/API and display integration	6	A5 FS, A6 FF	19	25	19	25	0	2023-10-12	2023-11-22
A10	System integration testing	6	A7 FS, A8 FS, A9 FS	25	31	25	31	0	2023-11-23	2024-01-03
A11	Security, privacy, and usability validation	5	A10 SS, A10 FF	26	31	26	31	0	2023-11-30	2024-01-03
A12	Pilot deployment preparation	4	A10 FS, A11 FS	31	35	31	35	0	2024-01-04	2024-01-31
A13	Operations handover and support readiness	3	A12 SF	28	31	32	35	4	2023-12-14	2024-01-03
A14	Final release and project closure	2	A12 FS, A13 FS	35	37	35	37	0	2024-02-01	2024-02-14

Critical path: A1 -> A2 -> A4 -> A6 -> A9 -> A10 -> A11 -> A12 -> A14. The calculated project duration is 37 weeks, leaving schedule buffer inside the one-year project window.

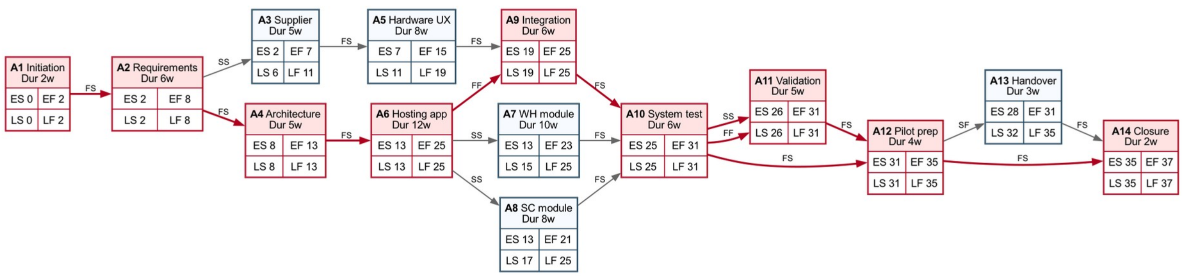

Figure 6: Activity network with ES/EF/LS/LF values. Red nodes are critical.

Gantt Chart

The Gantt chart translates the activity network into a calendar view from 2023-06-01. It shows the critical activities in red and non-critical activities in blue.

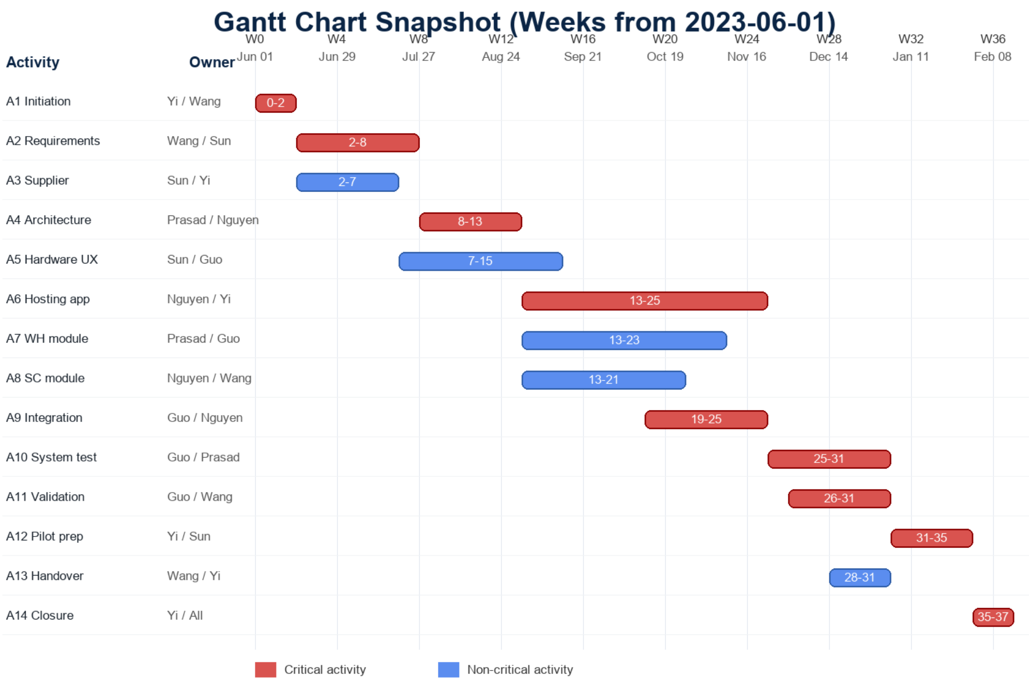

Figure 7: Gantt chart snapshot based on the activity network.

Resource and Skill Planning

Name	Working Time	PM	Req.	Arch.	Prog.	Security/Test	UX/Marketing	Docs
Hieu Nguyen	60%	3	3	4	5	4	2	3
Jingru Wang	50%	3	5	3	3	3	4	5
Prasad Dinesha Shyamala	60%	3	3	5	5	4	2	3
Jiahao Yi	60%	5	3	4	4	3	3	4
Teng Sun	50%	4	4	2	2	2	5	4
Ziyu Guo	50%	3	3	4	4	5	3	3

Activity	Primary Skills	Assigned Members	Capacity Note
A1	PM, stakeholder management	Yi, Wang	Setup, documentation, stakeholder records.
A2	Requirements, documentation	Wang, Sun, Yi	Requirements, market view, scope decisions.
A3	Supplier analysis, PM	Sun, Yi, Wang	Commercial criteria and decision traceability.
A4	Architecture, safety	Prasad, Nguyen, Guo	Architecture reviewed with security/testing view.
A5	Hardware/display, UX	Sun, Guo, Wang	Supplier data, usability, validation criteria.
A6	Programming, architecture	Nguyen, Yi, Prasad	Main hosting application development.
A7	Programming, UX	Wang, Nguyen	GUI and module configuration flows.
A8	Programming, integration	Prasad, Guo	WH integration and testing coverage.
A9	Programming, integration	Nguyen, Wang	SC cloud/location logic.
A10	Cloud/API integration	Guo, Nguyen, Prasad	Technical integration after supplier work.
A11	Testing, security, usability	Guo, Prasad, Wang	Usability, privacy, and security review.
A12	Deployment, documentation	Yi, Sun, Wang	Release package and handover documents.
A13	Operations readiness	Yi, Wang	Support readiness near pilot start.
A14	Closure	All members	Final review and project closure.

Conclusion

The iGEO project is feasible if TOMI treats safety separation, supplier maturity, stakeholder alignment, and privacy validation as core management topics. The optimized WBS contains twelve work packages, ProfCar is selected as the preferred supplier, the Use Case Point estimate gives approximately 121 UCP and 2,412 person-hours, and the activity network produces a 37-week critical-path schedule within the one-year constraint. The main residual risks are requirements volatility, supplier dependency, and user acceptance of the Smart Commercials module; these are addressed through early stakeholder involvement, controlled supplier selection, explicit validation, and a pilot release.